

An intraday trend-following trading strategy on equity derivatives in India

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In this article, we will present a trend-following based investment strategy on single-stock futures. Using the price movement of the recent past we are able to achieve a Sharpe Ratio of 1.7 on training data by cascading positions on successive positive signals and closing out positions if we hit a stop-loss. The stop-loss is computed using historical volatility. The universe is defined in an unbiased fashion to eliminate overfitting. We have used the top 75% most active single stock futures contracts and we have further filtered out products with low opening 15-minute volume. This was done to improve the scalability of the strategy. This might also help in avoiding contracts where less volatility is expected. We have trained our strategy on historical data from 2012 to 2016 and we have tested the strategy on the data from 2017 to 2018 data. Given the nature of markets in 2017-18, we have also looked at modifying the strategy to take positions more conservatively to avoid volatile situations. Empirical studies on profitable trading strategies are rare. We endeavor to shed light on the process of development of a profitable intraday trading strategy and we hope that this encourages collaboration in the active trading community.

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Introduction

Time Series Momentum is perhaps the most common and easily understood of the technical indicators. Because of its widespread popularity, it is arguably the most exploited yet a very consistent indicator present in equity markets. In this paper, we try to model a momentum indicator based on historical closing price action. This approach might be considered simplistic, but we show here that it leads to a profitable trading strategy. The performance of the strategy is further enhanced by employing volatility filters and volume filters to reduce the set of futures contracts we are taking positions on.

To reduce the mean reversion risk at the beginning, we employ a cascading strategy to build on an existing position if the indicator signal is still strong. We employ a trailing stop loss based approach using historical price deviation, current position and the maximum/minimum price achieved to find levels of retracement.

We find that the strategy performs really well for the period Jan-2012 to Dec-2015 but has its ups and downs from Jan-2016 to Dec-2018. This can be attributed to reduced persistence of time-series momentum in the latter period. We try to address this by a smarter cascading condition and a more efficient and sometimes profit-booking type stop-loss. These improvements don't affect the results too much for the period Jan-2012 to Dec-2016 but significantly boost the performance for the out-sample period Jan-2017 to Dec-2018.

Methodology

The underlying idea is to use historical price movement (mean, standard deviation) to find a threshold of price movement, crossing which, we take a position in the futures contracts in the direction of the price movement. For this, we employ trigger based momentum indicators.

We use historical (close price - previous close price) data to compute weighted mean and standard deviation.

$$\alpha * \text{mean} + \beta * \text{std} < \text{abs}(\text{current close} - \text{previous day close}) \quad (1)$$

The current formulation works best amongst close-open, high-low, abs(low/high-close). Mean performs better than Median for the dataset at hand. After experimentation, we settled on a price lookback of 5 business days i.e. 1 week, to compute the signal and granularity of 15 minutes to periodically check for the current closing price. Cascading on the signal after every favorable move significantly improves the alpha. Stop loss is placed in a trailing fashion which is a function of historical price deviation, current price, and position.

Support price = $f(\text{day high, std, \#lots})$

where #lots is the number of lots/cascades done, std is the same as (1).

The resistance price can be computed in a similar fashion. To improve the entry conditions, certain filters are placed to weed potentially weak signals. They are:

1. Volume filter

We compute the median of opening 15-minute volume for the last 180 days and allow trading if the opening 15-minute volume is greater than the median mentioned.

2. Volatility filter

- a. Price action: Compute the mean of the percentage returns of the product for the last 30 days and filter out the bottom 25% products (i.e. the ones with the lowest movement).
- b. Modified Volume Price Trend: For each product, we compute the summation of net price movement multiplied by volume for every granular bar. We normalize it by dividing by the total volume and price. Then we compute the standard deviation over the last 30 days. The bottom 25% of products are filtered out.

Investment Universe

The universe for trading is all single stock futures listed in NSE. We filter out 25% of the least volatile products by the methods mentioned under Volatility filters subsection below.

Portfolio Construction

1. To limit exposure per product, we have defined 1 lot as a fixed notional of 5 lac INR regardless of the product price.
2. The maximum exposure in each product can be of 10 lots, i.e., 50 lac INR.
3. With starting AUM of 10 cr, the average daily (90 percentile) exposure comes out to be 24.6%, i.e., 2.46 cr which is effectively 50 lots.

Data and Trading

To train and test our model, we use data from National Stock Exchange (NSE) and compress the trade information into minutely bars in open, close, low, high, volume (OCLHV) format. We have traded the strategy on NSE as well to get real execution statistics. Hence the execution statistics are based on real trading and not a theoretical model.

Experiments & Results

1. Cascade contributions:

The model mentioned above performs quite well for the in-sample (2012-16) in NSE futures market but underperforms in the out-sample (2017-18) because of increased volatility and a possible regime change. As an experiment to see the contribution from each cascade (enter at that price and hold until stop loss/market close), we see that there is a significant contribution from the 10 cascades in in-sample. However, the contribution from the first cascade in 2017-18 is significantly negative which means the presence of mean reversion and/or a weak momentum signal at the time of entry. Realizing the need for the strategy to be more conservative, we skip the first cascade and take a contribution from the second which significantly improves the results in 2017-18 without affecting 2012-16 much.

Out-sample cascade results

Cascade	1	2	3	4	5	6	7	8	9	10
Base	-0.15	0.16	0.14	0.18	0.18	0.16	0.13	0.1	0.07	0.04
Improved	0	0.16	0.13	0.16	0.16	0.13	0.11	0.08	0.05	0.02

Table 1: Here, the numbers are fractional contributions to the overall PnL. Results improve upon cascading entry into the position. The biggest impact is in changing the cascade number to around 6. However, there is a decent positive contribution to return all the way up to 8 or so.

Out-sample metrics

Metrics	Pnl	Sharpe Ratio	Drawdown	Pnl/drawdown
Base	76.54L	1.03	34.31L	1.26
Improved	79.64L	1.53	23.78L	1.89

Table 2: There is a huge improvement in the Sharpe Ratio when we ignore the first cascade.

2. Volatility filters

To weed out non-profitable trades, we try to improve volatility filters. Instead of

using the standard deviation of price action to remove low deviation products, we replace it with the mean of price action based filtering.

The results have a small negative but insignificant impact on in-sample but significantly improve the results for out-sample.

Top 75%ile Mean((close- previous close)/previous close)₃₀

Secondly, we use the standard deviation of the volume price trend to filter low deviation products. We compute net price movement * volume in that bar and sum over the full day and normalize it with total volume and closing price.

Top 75% Stdev($\Sigma(\text{close} - \text{open}) * \text{volume}_{\text{bar}} / (\text{total volume} * \text{previous close})$)₃₀

Out-sample volatility results:

Metrics	Pnl	Sharpe Ratio	Drawdown	Pnl/drawdown
Base	19.52L	0.36	27.39L	0.38
Improved	46.16L	0.9	23.52L	1.04

Table 3: Improvements due to volatility filters

Cumulative log-return charts and performance statistics

In-sample Results:

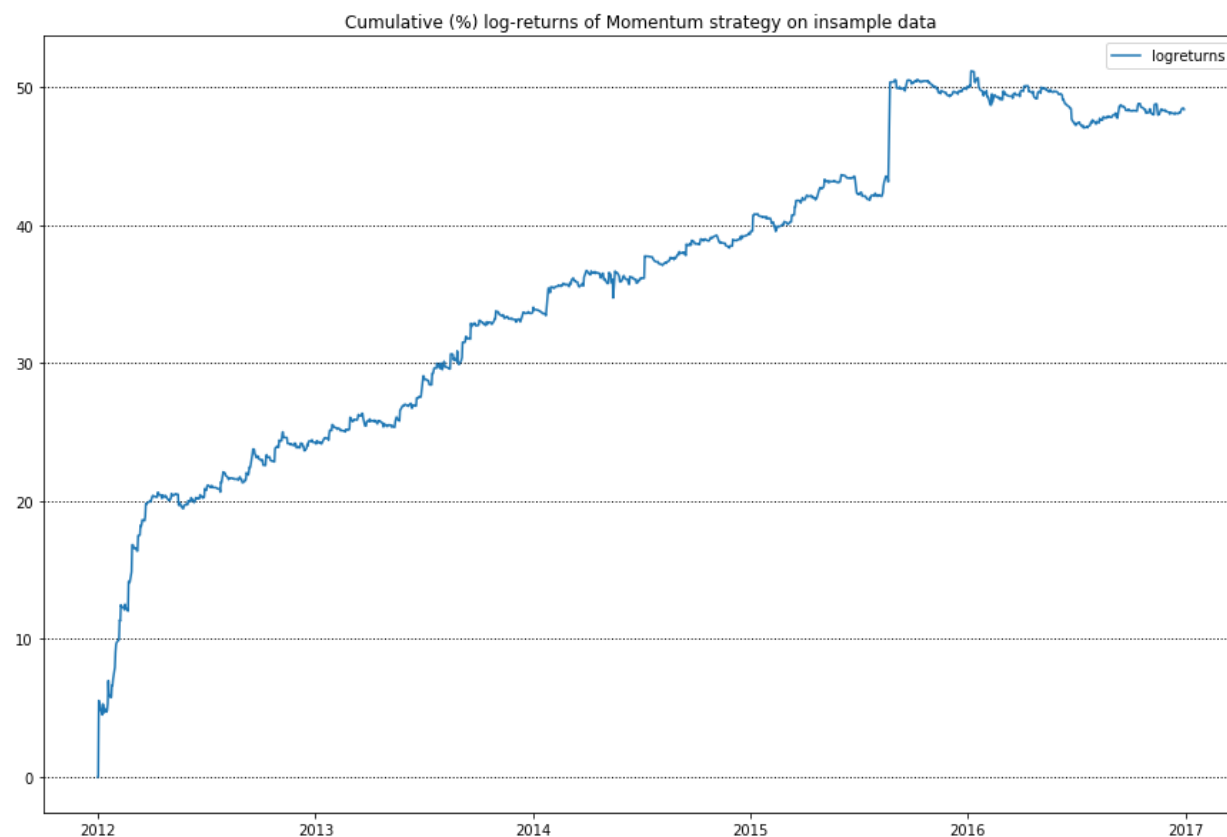


Figure 1: Cumulative log returns for in-sample period Jan-2012 to Dec-2016

In-sample - Time Series Momentum stats	
annualized_returns	9.78 %
annualized_stdev	5.56 %
net_percentage_returns	62.21 %
worst_drawdown	4.03 %
max_yearly_loss	2.72 %
return_drawdown_ratio	2.43
return_var10_ratio	34.90

omega_ratio	1.55
hit_loss_ratio	0.00
gain_pain_ratio	0.76
percent_positive_months	65.00 %
percent_positive_rolling_quarters	81.03 %
percent_positive_rolling_years	89.80 %
annual_sharpe_ratio_daily_returns	1.74
annual_sharpe_ratio_monthly_returns	1.44
standard_deviation_of_monthly_returns	6.74
sortino_ratio	5.30
months_to_recovery_from_max_dd	5
information_ratio	1.71

Table 4: Metrics for in-sample performance. The in-sample period is 2012-2016

Outsample:

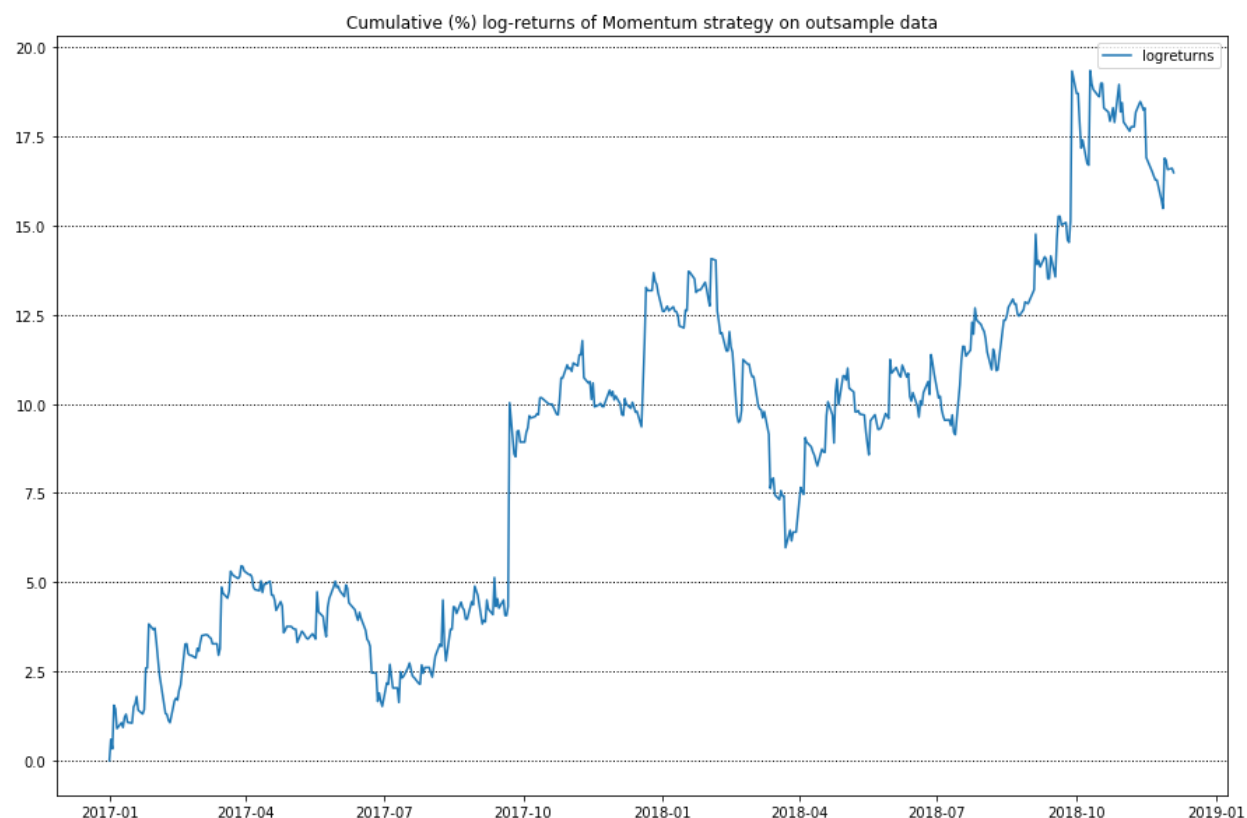


Figure 2: Cumulative log returns for in-sample period Jan-2017 to Dec-2018

Outsample - Time Series Momentum stats	
annualized_returns	8.61 %
annualized_stdev	9.01 %
net_percentage_returns	17.92 %
worst_drawdown	7.78 %
max_yearly_loss	0.82 %
return_drawdown_ratio	1.11
return_var10_ratio	11.72
omega_ratio	1.20

hit_loss_ratio	0.00
gain_pain_ratio	0.23
percent_positive_months	54.17 %
percent_positive_rolling_quarters	72.73 %
percent_positive_rolling_years	100.00 %
annual_sharpe_ratio_daily_returns	0.80
annual_sharpe_ratio_monthly_returns	0.83
standard_deviation_of_monthly_returns	8.70
sortino_ratio	1.56
months_to_recovery_from_max_dd	5
information_ratio	0.91

Table 5: Metrics for out-sample performance. Outsample period is 2017-2018

Execution sensitivity

1. We have tried to estimate execution cost from real trading of single stock futures. Although the cost varies for each product, they average at around 7 bps. Refer Table 6 below for some examples.
2. The total traded value is 2870.42cr for 2012-2016 with slippage amounting to 2cr and net profit (after accounting for slippage) is 2.045cr. So the strategy is earning a net of 7 bps on top of slippage assumption of 7bps.

Future Work

Execution improvement will contribute significantly to performance improvement. As we have mentioned in the section "Execution sensitivity", execution costs can be as significant as about 50% of the gross profits. A lot of future work needs to be focused on reducing execution costs.

Strategy improvement:

While the current momentum alpha has been significantly pruned to result in a robust indicator, there are always possible improvements to enhance the performance.

1. Analyze behavior and correlation with another trend based strategy in our portfolio which employs MACD and Bollinger to decide on the best portfolio allocation.
2. Expand the idea to trade and hold for probably longer periods and study its effects.
3. Possible enhancement can be achieved by employing machine learning to improve the underlying signal

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